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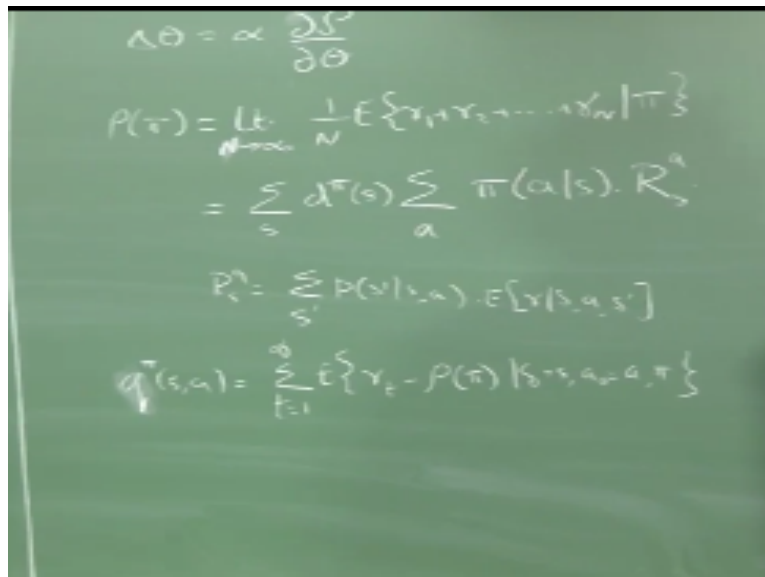
Proof of Theorem I

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So if you remember the way we defined our delta θ .

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$$\begin{aligned}\Delta \theta &= \alpha \frac{\partial J}{\partial \theta} \\ P(\pi) &= \lim_{N \rightarrow \infty} \frac{1}{N} E \{ r_1 + r_2 + \dots + r_N | \pi \} \\ &= \sum_s d^\pi(s) \sum_a \pi(a|s) \cdot R_s^a \\ R_s^a &= \sum_{s'} P(s'|s,a) \cdot E \{ r | s, a, s' \} \\ q^\pi(s,a) &= \sum_{t=1}^{\infty} E \{ r_t - P(\pi) | s=s, a=a, \pi \}\end{aligned}$$

Was going to be some α times, its some τ ρ by τ θ right or τ η by we defined it as τ η by τ θ where η was some performance measure right so one popular performance measure that we could have is essentially the average reward which is yes, so limit n tending to infinity of the sum of r_1 to r_N right we did this last time I put the N within the expectation right now I have taken the N outside the expectation otherwise it is exactly the same expression I wrote in the last class right for this is the average reward right.

So one way of thinking about the average reward is that it is given by the following where $R_s a$ is the reward you get for taking action a in state s right, so what have I done here I have marginalized over s' I just summed over all s' that you could land up just to make it easy, you could always write an extra summation here over s' and keep carrying it around so $R_s a$ is essentially, ok using our old notation.

Now this is what $R_s a$ is if you want you can write the summation inside that and you can keep carrying it around I just want to simplify this I put in a new notation for the summation okay so $R_s a$ is that and I am also going to define without giving you any reasons for it a new Q function $Q_\pi s a$ for the average reward case okay, which tells me if I start by taking action a , right if I start by taking action a so how much more than the average reward will I expect to get right.

It should not really be a difference if you think about it if I do not if I remove this difference and take the summation in right the $\sum_{t=1}^{\infty} \gamma^t r_t$ equal 1 to infinity $\sum_{t=1}^{\infty} \gamma^t \pi_t$ equals one to infinity should be, looks like it should be the same right so looks little counter intuitive I am just pointing it out to you already right but then this is called the this is called the ρ is typically called the gain of a policy right and this Q function is called the bias of a policy okay I think I got it right I think that is the gain and this is the bias right and this can actually be used to compare different actions right.

So if I start off by taking this action so there will be some initial transient terms that might not get cancelled out so that will tell me that I am slightly better than the other thing so this is why I said we need to do the whole couple of days of average reward RL so that you can understand what the difference between the gain and the bias and other things but right now I am defining this without any further explanation.

So let us take this as the Q function definition for average reward RL, this essentially tells you how much starting off with action a let us say a_1 is better than starting off with action a_2 , it is more of a comparative value function that you should be using why not, why, how will how will that happen.

Because this is π anyway I am behaving according to π I am just starting off with a after that I am behaving according to π right so the long term behavior for r_t should, should be tending towards ρ π right I mean sometimes it is high, much above ρ π sometimes it should be much below ρ π otherwise ρ π cannot be the average reward right you after the first action I am always behaving according to π right so it cannot be the average reward it is only the first action that is different right so if the first action gives me a slight advantage then I am going to say that it is a better action than starting off with something else okay.

It is essentially used as a comparative measure between actions τ so it make sense great so likewise we could define ρ for the discounted reward case also right I am if you remember if I am going to use discounted rewards what was the assumption I made for the policy gradient case what is the assumption we made, s_0 , right there is one unique s_0 and I take the value of s_0 as my value for the policy right.

So likewise I can define the same thing I can look at the expected discounted reward starting from s_0 , keep that as ρ and I can define my Q in the usual way starting from s_0 doing action a following policy π thereafter okay so that is my Q function and I need τ ρ by τ θ right so let us see what τ ρ by τ θ should be.

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$$\begin{aligned}
\frac{\partial J}{\partial \theta} &= \sum_s d^{\pi}(s) \sum_a \frac{\partial \pi(s,a)}{\partial \theta} q_v^{\pi}(s,a) \\
\frac{\partial J^{\pi}(s)}{\partial \theta} &= \frac{\partial}{\partial \theta} \sum_a \pi(s,a) \cdot q_v^{\pi}(s,a) \\
&= \sum_a \left[\frac{\partial \pi(s,a)}{\partial \theta} q_v^{\pi}(s,a) + \pi(s,a) \frac{\partial q_v^{\pi}(s,a)}{\partial \theta} \right] \\
&= \sum_a \left[\frac{\partial \pi(s,a)}{\partial \theta} \left(R_s^a - J^{\pi} \right) + \pi(s,a) \left(-\frac{\partial J^{\pi}}{\partial \theta} + \sum_{s'} P(s'|s,a) \frac{\partial J^{\pi}(s')}{\partial \theta} \right) \right] \\
\therefore \frac{\partial J}{\partial \theta} &= \sum_a \left[\frac{\partial \pi(s,a)}{\partial \theta} \left(R_s^a - J^{\pi} + \sum_{s'} P(s'|s,a) J^{\pi}(s') - J^{\pi} \right) \right]
\end{aligned}$$

This was a pretty amazing result so if you think about it, the π depends on θ right, all of you know that is why I have the τ π by τ θ but $d \pi$ also depends on π , so that should have a π that should also figure in the derivative right q π s a also depends on π right that should also figure in the derivative so if you just think about applying the derivative directly to this expression and essentially you will have to do it through the product rule right so there are three terms here and I should differentiate everything but then it turns out that that is not really the case right and the derivative of J by θ is you just basically have to take the derivative of the policy with respect to θ evaluated at a right.

And basically you are done ok this is an amazing result right, so there are variations of this results already in the literature but then this guy Peter Marbach came up with a proof for this right it runs into several pages maybe 15 pages or 20 pages or something of his thesis the proof of this theorem runs into several pages in his in his thesis right and then a little later without really knowing about Marbach's result before they were set out to prove these things.

So Rich Sutton Satinder Singh David McAllester and a bunch of people wrote this paper and that is the proof okay so Marbach's proof runs for several pages because he was working from a different set of initial principles and so Rich came up with this proof where he starts off with an expression that is completely unrelated to what they are trying to prove right, and then

somewhere along the line suddenly like a magician he derives his relation okay so I will show you this proof just because it is so elegant and so simple and so all of you know what the expression for V_{π} is do you?

I am using ρ of a given s and π_s , a as the same okay, so I can take the derivative of this right so what will this be right yes that is rather straight forward okay then what we do okay, so do we need to do anything to this first expression not really that is what we want to be left behind right so we will just leave it as it is right we will just leave the first expression as it is and we will play around with the second expression and see if we can make it, go away, so why is it $R_s a$.

So the expected value of $r_t - \rho \pi_t$, the expected value of r_t is essentially $R_s a$, right where did we write $R_s a$, so this is $R_s a$ right so that is the expected value I will get for the first step for taking action a right, so expected value of r_t will be $R_s a$, then $-\rho \pi_t$ and then what do we have so why π_t that come, right t equal to 1 to infinity, so this is like the first term right so this is the rest of the terms whatever follows later I am writing it in terms of the value function right.

So this will go away because π_t not depend on θ right so you will get - okay finally we got our $\tau \rho$ by $\tau \theta$ so far we did not have that expression at all right what we wanted to derive $\tau \rho$ by $\tau \theta$, so I did not start from this expression I am sorry I did not start from where is the ρ expression this expression I did not take the derivative right I started from some arbitrary equation this is v_{π} equal Q_{π} right and then I started making substitutions and then I took the derivatives I finally got my term $\tau \rho$ by $\tau \theta$.

Now I will have to rewrite it so that this comes to the left hand side ok so what do I get therefore right so I think that is correct I did nothing I just took the $\tau \rho$ by $\tau \theta$ to the other side so if you think about it I multiply this by $\pi_s a$ right but I can pull it out because it is not dependent on a and then sum overall a so that $\pi_s a$ will go to one, right so I will be left with $-\tau \rho$ by $\tau \theta$ and bring $-\tau \rho$ by $\tau \theta$ here, I take τv_{π} by $\tau \theta$ that side right and the rest of the terms I have written it like this okay is it clear okay.

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$$\begin{aligned}
 \sum_s d^T(s) \frac{\partial J^*(s)}{\partial \theta} &= \sum_s d^T(s) \sum_a \frac{\partial \pi(a|s)}{\partial \theta} \cdot q^T(s,a) \\
 &\quad + \sum_s d^T(s) \sum_a \pi(a|s) \sum_{s'} p(s'|s,a) \frac{\partial J^*(s')}{\partial \theta} \\
 &\quad - \sum_s d^T(s) \frac{\partial J^*(s)}{\partial \theta} \\
 &\rightarrow \sum_s \sum_a \sum_{s'} d^T(s) \pi(a|s) \cdot p(s'|s,a) \frac{\partial J^*(s')}{\partial \theta} \\
 &\quad - \sum_{s'} d^T(s') \frac{\partial J^*(s')}{\partial \theta}
 \end{aligned}$$

So more sleight-of-hand now right so I am going to take expectations with respect to d π s on both sides and I am going to take expectation with respect to deep is on both sides so essentially what that would be $\sum_s d$ π s into ρ of $\pi = \sum_s d$ π s into $\sum_a$, I have written it correctly, right so the first term $\sum d$ π s, the second term d π s, the third term d π s, okay, where τ , this is fine right, so now comes the amazing trick so we know that d π s is a stationary distribution right so let me just take this term alone right and write it like this.

Right so what τ s this mean ok I am going to start off with the stationary distribution s , I mean d π at some state s I will take an action according to my policy π then I will make a transition according to p this whole thing what do I expect this distribution to be like because d π is the steady state distribution the condition of steady state distribution is it should be the same right the condition when do you call d π the steady state distribution if I apply π and p on it I should get back the same distribution right only then it is a steady state distribution so that is essentially what is happening it right if you think across all this $\sum s$ prime and a , essentially I have applied π and p on d π right if I evaluate these two summations over this I will essentially end up with d π of s prime right.

So this is essentially equal to $\sum s$ prime okay, this looks familiar, of course I can ignore this because there is the probability distribution it is one right so I can ignore this so basically I have my theorem $\tau \rho$ by $\tau \theta = \sum s, d \pi s$ into a $\tau \pi s \sum a \tau \theta$ into $Q \pi s a$, amazing right, whatever possessed them to start off with the equivalence between v and Q to get this derivation on ρ right it never ceases to amaze me that the Rich managed to find something like this so what is the nice thing is it gives you a good connection between using value functions and, so what is the difference between what we are doing now and what the motivation I gave you earlier, in the motivation I gave you earlier you ended up using the v function.

v was your baseline right when we said actor critic so the actor was the policy but the critic was $v(s)$, here the critic you are using is $Q s a$ right and the amazing thing is it turns out the same form holds regardless of whether you are using the average reward formulation like we did now or whether you are using the discounted reward formulation except that you need to do a different sleight of hand to get the results but the proof holds whether you are using average reward formulation or whether you are using discounted reward formulation right.

I encourage you to read the proof for the discounted reward formulation so I am without a TA, so you are my TA so just remind the TAs to put up this Sutton McAllester Singh and Mansour paper online so I encourage people to read this, this is actually a really old paper I think is from 99 or something but still it is amazingly instructive as to how you can think out of the box to get really simple proofs for things okay this is one thing now it has now connected policy gradient approaches to the q value function okay and it is still not done.

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